The Binomial Expansion

Starters

Solution 2

1. Expand and simplify.

a $(2x-3y)^2$

a
$4x^2-12xy+9y^2$

b $(x-y)^3$

b
$x^3-3x^2y+3xy^2-y^3$

c $(2+x)^3$

c
$8+12x+6x^2+x^3$

2. Simplify.

a $(-2x)^3$

a
$-8x^3$

b $(3x)^{-4}$

b
$\frac{1}{81x^4}$

c $(\frac{2}{5}x)^2$

c
$\frac{4}{25}x^2$

d $(\frac{1}{3}x)^{-3}$

d
$\frac{27}{x^3}$

3. Simplify using root notation.

a $(25x)^{1/2}$

a
$5\sqrt{x}$

b $(64x)^{-2/3}$

b
$\frac{1}{16\sqrt[3]{x^2}}$

c $(\frac{9}{100}x)^{-1/2}$

c
$\frac{10}{3\sqrt{x}}$

d $(\frac{8}{27}x)^{4/3}$

d
$\frac{16}{81}\sqrt[3]{x^4}$

Chapter Overview

Solution 3

1. Pascal's triangle

Recognise the coefficients in binomial expansions.

2. Factorial and choose notation

Calculate coefficients directly.

3. Binomial expansion

Expand brackets and find a specified term.

4. Estimation

Use the first few terms to approximate powers.

Starter

Solution 4

0 Expand $(a+b)^0$

0

1

1 Expand $(a+b)^1$

1

$a+b$

2 Expand $(a+b)^2$

2

$a^2+2ab+b^2$

3 Expand $(a+b)^3$

3

$a^3+3a^2b+3ab^2+b^3$

4 Expand $(a+b)^4$

4

$a^4+4a^3b+6a^2b^2+4ab^3+b^4$

Coefficients The coefficients:

Coefficients

They follow Pascal's triangle (we'll explore this on the next slide).

Powers The powers of a and b:

Powers

Power of a decreases each time (starting at the power).

Power of b increases each time (starting at 0).

More on Pascal's Triangle

Solution 5

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
1   5   10  10  5   1
```

Each interior entry is the sum of the two entries above it. Each row starts and ends with 1.

Number the top row 0. Row n supplies the coefficients of $(a+b)^n$. Learn rows 0 to 5.

Example Which row is used for $(2+x)^4$?

Example

1, 4, 6, 4, 1

Example

Solution 6

Find the expansion of $(2+3x)^4$.

Use descending powers of the first term and ascending powers of the second. The powers add to the power of the bracket. A term raised to power 0 is 1.

Coefficients First fill in the correct row of Pascal's triangle.

Coefficients 1, 4, 6, 4, 1

Term 1

$1(2^4)$

Term 2

$4(2^3)(3x)$

Term 3

$6(2^2)(3x)^2$

Term 4

$4(2)(3x)^3$

Term 5

$1(3x)^4$

Result Simplify each term, raising every number in a bracket to the appropriate power.

Result $(2+3x)^4 = 16 + 96x + 216x^2 + 216x^3 + 81x^4$

Initially write one line per term before simplifying. This gives each term enough space.

Another Example

Solution 7

Find the expansion of $(1-2x)^3$.

Use descending powers of the first term and ascending powers of the second. The powers add to the power of the bracket. A term raised to power 0 is 1.

Coefficients First fill in the correct row of Pascal's triangle.

Coefficients $1, 3, 3, 1$

Term 1

$1(1^3)$

Term 2

$3(1^2)(-2x)$

Term 3

$3(1)(-2x)^2$

Term 4

$1(-2x)^3$

Result Simplify each term, raising every number in a bracket to the appropriate power.

Result $(1-2x)^3 = 1 - 6x + 12x^2 - 8x^3$

$(1-2x) = (1+(-2x))$, so use $-2x$ as the second term. The signs alternate positive and negative; check this when simplifying.

Getting a single term in the expansion

Solution 8

The coefficient of x^2 in the expansion of $(2-cx)^5$ is 720. Find the possible values of the constant c.

The "5" row in Pascal's triangle is 1, 5, 10, 10, 5, 1. Counting the first entry as the 0th term, the coefficient we want is 10.

1 • The power of $(-cx)$ must be

1

2.

2 • The power of 2 must be

2

3 (the two powers must add up to 5).

3 Therefore the term is:

3

$10(-cx)^2(2^3) = 80c^2x^2$

$80c^2 = 720, \qquad c^2 = 9, \qquad c = \pm 3$

Test Your Understanding — Solution 9

a Find the first 3 terms, in ascending powers of x, of $(2+kx)^7$, where k is a constant. Give each term in its simplest form.

a

$$(2+kx)^7 = 2^7 + 7(2^6)(kx) + \binom{7}{2}(2^5)(kx)^2 + \ldots$$
$$= 128 + 448kx + 672k^2x^2 + \ldots$$

b Given that the coefficient of x^2 is 6 times the coefficient of x, find k.

b

$672k^2 = 6(448k)$

$672k(k-4) = 0$

$k = 0$ or $k = 4$

The wording does not exclude zero. If non-zero coefficients are intended, k=4.

Exercise 8A – Y1 page 160-161 — Solution 10

4 Fully expand the expression $(1+3x)(1+2x)^3$.

4

$(1+2x)^3 = 1 + 6x + 12x^2 + 8x^3$

$(1+3x)(1+2x)^3$

$= 1 + 9x + 30x^2 + 44x^3 + 24x^4$

5 Expand $(2+y)^3$. Hence or otherwise, write down the expansion of $(2+x-x^2)^3$ in ascending powers of x.

5

$(2+y)^3 = 8 + 12y + 6y^2 + y^3$

Substitute $y=x-x^2$:

$(2+x-x^2)^3 = 8 + 12(x-x^2) + 6(x-x^2)^2 + (x-x^2)^3$

$= 8 + 12x - 6x^2 - 11x^3 + 3x^4 + 3x^5 - x^6$

Exercise 8A – Y1 page 160-161 Solution 11

6 The coefficient of x^2 in the expansion of $(2+ax)^3$ is 54. Find the possible values of the constant a.

6

$[x^2](2+ax)^3 = 3 \cdot 2(ax)^2/x^2 = 6a^2$

$6a^2 = 54 \Rightarrow a = \pm 3$

7 The coefficient of x^3 in the expansion of $(2-x)(3+bx)^3$ is 45. Find possible values of the constant b.

7

$(3+bx)^3 = 27 + 27bx + 9b^2x^2 + b^3x^3$

$2b^3 - 9b^2 = 45$

$2b^3 - 9b^2 - 45 = 0$

A numerical solver gives the single real solution b≈5.3008.

Exercise 8A – Y1 page 160-161 Solution 12

8 Work out the coefficient of x^2 in the expansion of $(p-2x)^3$. Give your answer in terms of p.

8

The coefficients are $1, 3, \mathbf{3}, 1$.

The term in x^2 is $3(p)(-2x)^2 = 12px^2$.

So the coefficient of the term x^2 is $12p$.

9 After 5 years, the value of an investment of £500 at an interest rate of $X\%$ per annum is given by $500(1+X/100)^5$

Find an approximation for this expression in the form $A + BX + CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

9

$500(1+X/100)^5$

$\approx 500(1 + 5X/100 + 10X^2/10000)$

$= 500 + 25X + \frac{1}{2}X^2$

$A = 500, \quad B = 25, \quad C = \frac{1}{2}$

Extension

Solution 13

MAT 2009 1J How many pairs of positive integers x,y satisfy

$x^3 + 6x^2y + 12xy^2 + 8y^3 = 2^{30}$

A: 0 B: 2^6 C: $2^9 - 1$ D: $2^{10} + 2$

MAT 2009 1J

$(x + 2y)^3 = 2^{30} \Rightarrow x + 2y = 2^{10}$

$x = 2(512 - y)$

For x,y positive integers, y=1,2,. . .,511. Each gives exactly one x. There are $511=2^9-1$ pairs: C.

Factorial and Choose Function

Solution 14

Factorial

$n! = n(n-1)\cdots 2 \cdot 1$

The number of arrangements of n distinct objects in a line. Also 0!=1.

For A,B,C, there are 3 choices, then 2, then 1:

$3! = 6$

Use your calculator's factorial function.

Choose

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

The number of selections of r objects from n distinct objects, when order does not matter.

Choose 4 players from 10:

$$\binom{10}{4} = \frac{10!}{4!6!} = 210$$

Use your calculator's nCr function.

Examples

Solution 15

Calculate the following. You may use the factorial button, but not the nCr button.

a $5!$

a $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

b $\binom{5}{3}$

b $\binom{5}{3} = \frac{5!}{3!2!} = \frac{120}{6 \times 2} = 10$

c $0!$

c
$0! = 1$. Accept this for now; part (e) explains why.

d $\binom{20}{1}$

d
There are 20 ways to choose 1 object from 20.
$\binom{20}{1} = \frac{20!}{1!19!} = 20$. In general, $\binom{n}{1} = n$.

e $\binom{20}{0}$

e
Choosing no objects is one possible selection.
$\binom{20}{0} = \frac{20!}{0!20!} = 1$, which requires $0! = 1$. In general, $\binom{n}{0} = 1$.

f $\binom{20}{2}$

f
$$\binom{20}{2} = \frac{20!}{2!18!} = \frac{20 \times 19}{2!} = 190.$$
In general, $\binom{n}{2} = \frac{n(n-1)}{2}$.

g $\binom{20}{18}$

g
$\binom{20}{18} = \frac{20!}{18!2!} = \binom{20}{2} = 190.$
When the bottom number exceeds half the top, subtract it from the top to simplify.

Why do we care?

Solution 16

For large powers such as $(x+3)^{20}$, building Pascal's triangle becomes impractical. The choose function gives any entry directly.

n	row entries
0	1
1	1 1
2	1 2 1
3	1 3 3 1
4	1 4 6 4 1

$\binom{4}{0} = 1, \quad \binom{4}{1} = 4, \quad \binom{4}{2} = 6$
$\binom{4}{3} = 4, \quad \binom{4}{4} = 1$

Number the top row 0 and the first entry 0. Entry k in row n is $\binom{n}{k}$: the top number gives the row, the bottom number gives the entry.

The textbook starts counting at 1; here we count from 0 so the numbers match the binomial coefficient directly.

The triangle is symmetric:

$$\binom{4}{1} = \binom{4}{3}.$$

Choosing one object is equivalent to choosing the three objects to leave out.

Interesting facts (not required for A level) — Solution 17

Pascal's rule: each interior entry is the sum of the two above it.

$$\binom{3}{1}+\binom{3}{2}=3+3=6=\binom{4}{2}$$

$$\binom{n-1}{k-1}+\binom{n-1}{k}=\binom{n}{k}$$

To choose k objects from n, focus on one particular object. Either include it and choose k−1 from the other n−1, or exclude it and choose k from the other n−1. These cases are disjoint and cover every selection.

Exercise 8B – Y1 page 162 — Solution 18

10 Given that $\binom{50}{13}=\frac{50!}{13!a!}$, write down the value of a.

10

$\binom{50}{13}=\frac{50!}{13!(50-13)!}=\frac{50!}{13!37!}$.

$a=37$.

11 Given that $\binom{35}{17}=\frac{35!}{p!18!}$, write down the value of p.

11

$\binom{35}{17}=\frac{35!}{17!(35-17)!}=\frac{35!}{17!18!}$.

$p=17$.

Exercise 8B – Y1 page 162 — Solution 19

a Work out $\binom{10}{3}$ and $\binom{10}{7}$.

b Work out $\binom{14}{5}$ and $\binom{14}{9}$.

a

$\binom{10}{3} = \frac{10!}{3!7!} = 120, \qquad \binom{10}{7} = 120$

b

$\binom{14}{5} = \frac{14!}{5!9!} = 2002, \qquad \binom{14}{9} = 2002$

c What do you notice about your answers to parts a and b?

c

The two answers in part a are equal, and the two answers in part b are equal.

d Prove that $\binom{n}{r} = \binom{n}{n-r}$.

d

$$\binom{n}{n-r} = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}$$

Using Binomial Coefficients to Expand — Solution 20

Why do binomial coefficients appear?

$(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$

Every product term chooses one item from each bracket.

Count How many selections produce a^3b^2?

Count

Choose which 3 of the 5 brackets supply a; the others supply b.

$$\binom{5}{3} = 10$$

Therefore the term is $10a^3b^2$.

The whole coefficient row is:

$\binom{5}{0}, \binom{5}{1}, \binom{5}{2}, \binom{5}{3}, \binom{5}{4}, \binom{5}{5}$

$1, 5, 10, 10, 5, 1$

Using Binomial Coefficients to Expand

Solution 21

For a non-negative integer n:

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

The powers in each term sum to n. Negative and fractional powers require the Year 2 generalised expansion.

Example Find the first four terms of $(3x+1)^{10}$ in ascending powers of x.

Example

$$(1+3x)^{10} = \sum_{r=0}^{10} \binom{10}{r} (3x)^r$$

$$= 1 + 30x + 405x^2 + 3240x^3 + \cdots$$

Test Your Understanding

Solution 22

Answer Find the first 3 terms in the expansion of $(2 - \frac{1}{3}x)^7$, in ascending powers of x.

Answer

$$2^7 + 7 \cdot 2^6(-x/3) + 21 \cdot 2^5(-x/3)^2 + \cdots$$

$$= 128 - \frac{448}{3}x + \frac{224}{3}x^2 + \cdots$$

Note: The "+ . . ." indicates that there would have been other terms in the expansion.

Exercise 8C – Y1 page 164 Solution 23

5 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2-x)^6$ and simplify each term.

5

$$(2-x)^6 = 2^6 + \binom{6}{1}2^5(-x) + \binom{6}{2}2^4(-x)^2 + \ldots$$
$$= 64 - 192x + 240x^2 + \ldots$$

6 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3-2x)^5$ giving each term in its simplest form.

6

$$(3-2x)^5 = 3^5 + \binom{5}{1}3^4(-2x) + \binom{5}{2}3^3(-2x)^2 + \ldots$$
$$= 243 - 810x + 1080x^2 + \ldots$$

7 Find the binomial expansion of $\left(x + \frac{1}{x}\right)^5$ giving each term in its simplest form.

7

The general term is

$$\binom{5}{r}x^{5-r}(x^{-1})^r = \binom{5}{r}x^{5-2r}$$

$$x^5 + 5x^3 + 10x + \frac{10}{x} + \frac{5}{x^3} + \frac{1}{x^5}$$

Extension Solution 24

1: AEA 2013 In $(1 + \frac{12n}{5}x)^n$, the coefficients of x^2 and x^3 are equal and non-zero. Find n.

1: AEA 2013

$$\binom{n}{2}\left(\frac{12n}{5}\right)^2 = \binom{n}{3}\left(\frac{12n}{5}\right)^3$$

$$15 = 12n(n-2) \quad \Rightarrow \quad 4n^2 - 8n - 5 = 0$$

$$(2n+1)(2n-5) = 0 \quad \Rightarrow \quad n = -\frac{1}{2} \text{ or } \frac{5}{2}$$

This extension uses the generalised binomial expansion for non-integer n, developed in Year 2.

2 [STEP I 2010 Q5a] By considering $(1+x)^n$, where n is a positive integer, or otherwise, show that:

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n} = 2^n.$$

2

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \ldots + \binom{n}{n}x^n.$$

Set $x = 1$: $\quad 2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n}.$

Getting a single term in the expansion

Solution 25

For $(a+b)^n$ the term with b^r is

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

Write the bracket in an order that makes the required power easy to select.

1 Find the x^3 term in $(a+x)^{10}$.

1

$$\binom{10}{3} a^7 x^3$$

2 Find the x^{50} term in $(2x-1)^{75}$.

2

$$\binom{75}{50} (-1)^{25} (2x)^{50}$$

3 Find the x^7 term in $(3-x)^{12}$.

3

$$\binom{12}{7} 3^5 (-x)^7$$

4 Find the x^3 term in $(3x+4)^{16}$.

4

$$\binom{16}{3} 4^{13} (3x)^3$$

Getting a single term in the expansion

Solution 26

The coefficient of x^4 in $(1+qx)^{10}$ is 3360. Find the possible values of q.

1 Term is:

1

$$\binom{10}{4} (1^6)(qx)^4 = 210q^4x^4$$

2 Therefore:

2

$$\begin{aligned} 210q^4 &= 3360 \\ q^4 &= 16 \\ q &= \pm 2 \end{aligned}$$

Test Your Understanding — Solution 27

Answer In $(1+ax)^{10}$, where $a \neq 0$, the coefficient of x^3 is double the coefficient of x^2. Find a.

Answer

x^2 term: $\binom{10}{2}(1^8)(ax)^2 = 45a^2x^2$

x^3 term: $\binom{10}{3}(1^7)(ax)^3 = 120a^3x^3$

$$120a^3 = 2(45a^2)$$

$$120a^3 = 90a^2$$

$$a^2(4a-3) = 0$$

$$a = 0 \text{ or } \frac{3}{4}$$

But $a \neq 0$, so $a = \frac{3}{4}$.

Exercise 8D – Y1 page 166-167 — Solution 28

5. When $(1-2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p > 0$, use this informationto find:

a the value of the constant p

b the coefficient of x

c the coefficient of x^3

a Find p.

a

$\binom{p}{2}(-2)^2 = 2p(p-1) = 40$

$p^2 - p - 20 = (p-5)(p+4) = 0$

Since p>0, p=5.

b Coefficient of x.

b

$-2p = -10$

c Coefficient of x^3.

c

$\binom{5}{3}(-2)^3 = 10(-8) = -80$

Exercise 8D – Y1 page 166-167 Solution 29

Question 6

a Find the first three terms, in ascending powers of x, of the binomial expansion of $(5+px)^{30}$, where p is a non-zero constant.

a

$$\begin{aligned}(5+px)^{30} &= 5^{30} + \binom{30}{1}5^{29}(px) + \binom{30}{2}5^{28}(px)^2 + \dots \\ &= 5^{30} + 5^{29}(30px) + 5^{28}(435p^2x^2) + \dots\end{aligned}$$

b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x, work out the value of p.

b

$$\begin{aligned}5^{28}(435p^2) &= 29 \times 5^{29}(30p) \\ 435p^2 &= 29 \times 5(30p) \\ 435p^2 &= 4350p \\ p^2 &= 10p \\ p^2 - 10p &= 0 \\ p(p-10) &= 0 \\ p = 0 &\text{ or } p = 10\end{aligned}$$

p is a non-zero constant, so $p = 10$.

Exercise 8D – Y1 page 166-167 Solution 30

Question 7

a Find the first four terms, in ascending powers of x, of the binomial expansion of $(1+qx)^{10}$, where q is a non-zero constant.

a

$$\begin{aligned}(1+qx)^{10} &= 1^{10} + \binom{10}{1}1^{9}(qx) + \binom{10}{2}1^{8}(qx)^2 + \binom{10}{3}1^{7}(qx)^3 + \dots \\ &= 1 + 10qx + 45q^2x^2 + 120q^3x^3 + \dots\end{aligned}$$

b Given that in the expansion of $(1+qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q.

b

Coefficient of x^3 is $120q^3$. Coefficient of x is $10q$.

$$\begin{aligned}\text{So } 120q^3 &= 108 \times 10q \\ \Rightarrow 120q^3 - 1080q &= 0 \\ \Rightarrow 120q(q^2-9) &= 0 \\ \Rightarrow 120q(q+3)(q-3) &= 0 \\ q = 0,\ q = -3 &\text{ or } q = 3\end{aligned}$$

But as q is non-zero, $q = \pm 3$.

Exercise 8D – Y1 page 166-167 Solution 31

Question 8

a Find the first three terms, in ascending powers of x, of the binomial expansion of $(1+px)^{11}$, where p is a constant.

a

$$(1+px)^{11} = 1^{11} + \binom{11}{1}1^{10}(px) + \binom{11}{2}1^{9}(px)^2 + \dots$$
$$= 1 + 11px + 55p^2x^2 + \dots$$

b The first 3 terms in the same expansion are $1, 77x$ and qx^2, where q is a constant. Find the value of p and the value of q.

b

$11p = 77$ and $55p^2 = q$.

$p = 7$.

$q = 55 \times 7^2 = 2695$.

$p = 7, q = 2695$.

Exercise 8D – Y1 page 166-167 Solution 32

Question 9

a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1+px)^{15}$, where p is a non-zero constant.

a

$1 + 15(px) + 105(px)^2 + \cdots$

$= 1 + 15px + 105p^2x^2 + \cdots$

b Given that, in the expansion of $(1+px)^{15}$, the coefficient of x is $-q$ and the coefficient of x^2 is $5q$, find the value of p and the value of q.

b

$15p = -q, \qquad 105p^2 = 5q$

$21p^2 = q = -15p$

$3p(7p+5) = 0$

Since p is non-zero:

$p = -\frac{5}{7}$

$q = -15p = \frac{75}{7}$

Exercise 8D – Y1 page 166-167 Solution 33

10 In the binomial expansion of $(1+x)^{30}$, the coefficients of x^9 and x^{10} are p and q respectively. Find the value of $\frac{q}{p}$.

10

$$p = \binom{30}{9}, \quad q = \binom{30}{10}$$

$$\frac{q}{p} = \frac{30!}{10!20!} \cdot \frac{9!21!}{30!}$$

$$= \frac{21}{10} = 2.1 \quad \text{exactly}$$

Extension Solution 34

1: MAT 2014 For positive integer n, find the coefficient of x^3y^5 in $(1+xy+y^2)^n$.

A: n B: 2^n C: $\binom{n}{3}\binom{n}{5}$

D: $4\binom{n}{4}$ E: $\binom{n}{8}$

1: MAT 2014

Choose xy from 3 brackets, y^2 from one further bracket, and 1 from the rest.

$$\binom{n}{3}(n-3) = 4\binom{n}{4}$$

Answer D. For n<4 the coefficient is zero.

2: STEP I 2013 Compare coefficients of x^r in $(1+x)(1+x)^n$ to show

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$$

2: STEP I 2013

The product equals $(1+x)^{n+1}$, with coefficient

$$\binom{n+1}{r}$$

Multiplying out gives one contribution from 1 times x^r, and one from x times x^{r-1}:

$$\binom{n}{r} + \binom{n}{r-1}$$

Equating the coefficients proves the identity.

Estimating Powers — Solution 35

a Find the first 4 terms, in ascending powers of x, of $\left(1+\frac{x}{4}\right)^8$, giving each term in its simplest form.

a

$$\left(1+\frac{x}{4}\right)^8 = 1+8\left(\frac{x}{4}\right)+28\left(\frac{x}{4}\right)^2+56\left(\frac{x}{4}\right)^3+\ldots$$
$$= 1+2x+\frac{7}{4}x^2+\frac{7}{8}x^3+\ldots$$

b Use your expansion to estimate 1.025^8, giving your answer to 4 decimal places.

b

$1+\frac{x}{4}=1.025$, so $x=0.1$.

$$1.025^8 \approx 1+2(0.1)+\frac{7}{4}(0.1)^2+\frac{7}{8}(0.1)^3 = 1.2184 \quad (4\text{ d.p.}).$$

Why? Why should this be a reasonably good approximation despite the missing terms?

Why?

As the power increases, 0.1^r becomes very small. The omitted terms have negligible effect on this approximation.

Tip: compare against the calculator value of 1.025^8.

Test Your Understanding — Solution 36

(a) Find the first 4 terms of the expansion of $\left(1+\frac{x}{2}\right)^{10}$ in ascending powers of x, giving each term in its simplest form. (4)

(b) Use your expansion to estimate the value of $(1.005)^{10}$, giving your answer to 5 decimal places. (3)

a

$$\left(1+\frac{x}{2}\right)^{10} = 1+10\left(\frac{x}{2}\right)+45\left(\frac{x}{2}\right)^2+120\left(\frac{x}{2}\right)^3+\ldots$$
$$= 1+5x+\frac{45}{4}x^2+15x^3+\ldots$$

b

Use $x=0.01$, so $1+\frac{x}{2}=1.005$.

$$(1.005)^{10} \approx 1+5(0.01)+\frac{45}{4}(0.01)^2+15(0.01)^3$$
$$= 1+0.05+0.001125+0.000015$$
$$= 1.05114 \quad (5\text{ decimal places}).$$

Exercise 8E – Y1 Page 168-169 Solution 37

7

a. Find the first 3 terms, in ascending powers of x, of $\left(3-\frac{x}{5}\right)^{10}$, giving each term in its simplest form.

b. Explain how you would use your expansion to estimate 2.98^{10}.

a

$$\left(3-\frac{x}{5}\right)^{10} = 3^{10} + \binom{10}{1}3^9\left(-\frac{x}{5}\right) + \binom{10}{2}3^8\left(-\frac{x}{5}\right)^2 + \ldots$$
$$= 59049 - 39366x + 11809.8x^2 + \ldots$$

b

We need $3-\frac{x}{5} = 2.98$, so $x = 0.1$. Substitute this into the expansion from part a:

$2.98^{10} \approx 59049 - 39366(0.1) + 11809.8(0.1)^2$.

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8

a. Find the first 4 terms, in ascending powers of x, of $(1-3x)^5$.

b. Hence find a linear approximation to $(1+x)(1-3x)^5$ for small x.

a

$$(1-3x)^5 = 1^5 + \binom{5}{1}1^4(-3x) + \binom{5}{2}1^3(-3x)^2 + \binom{5}{3}1^2(-3x)^3 + \ldots$$
$$= 1 - 15x + 90x^2 - 270x^3 + \ldots$$

b

For the expansion $(1-3x)^5$, only use the first two terms as x^2 and higher powers can be ignored.

$$(1+x)(1-3x)^5 \approx (1+x)(1-15x)$$
$$\approx 1 - 15x + x - 15x^2$$
$$\approx 1 - 14x$$

9. A microchip company models the probability of having no faulty chips on a single production run as:

$P(\text{no fault}) = (1-p)^n, \quad p < 0.001,$

where p is the probability of a single chip being faulty, and n is the total number of chips produced.

a State why the model is restricted to small values of p.

b Given that $n = 200$, find an approximate expression for $P(\text{no fault})$ in the form $a + bp + cp^2$.

a

Small p makes truncation after p^2 useful for fixed n. The exact independent-chip probability $(1-p)^n$ does not itself require $p<0.001$.

b

$$(1-p)^{200}$$

$$\approx 1 - 200p + \binom{200}{2}p^2$$

$$= 1 - 200p + 19900p^2$$

c The company wants a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part b to suggest a maximum value of p for this to be the case.

c

$$19900p^2 - 200p + 0.08 = 0$$

$$p = \frac{200 \pm \sqrt{33632}}{39800}$$

The roots are approximately 0.00963 and 0.000417. Only the latter satisfies $p<0.001$, so the approximate maximum is $p=0.000417$ (3 s.f.).